

Movie-Level Joint Inference for Open-Ended Long-Video QA

Abstract. Open-ended long-video QA requires answers to remain consistent across a film’s characters, relationships, events, and ending state. We treat the movie as the inference unit: a single request combines timestamped subtitles, chronological visual observations, and the complete question set to generate one answer bank. The final SLoMO submission ranked first with 66.36% accuracy and a 3.02 score.

Keywords: long-video QA · movie-level inference · reproducibility

1 System Overview

The movie is the context unit. A shared request preserves aliases, relationships, event order, causes, and ending state across every question in the film. This organization aligns the reasoning context with the narrative scope of the task and yields one coherent answer bank.

2 Method

For movie m , shared evidence E_m and its complete question set Q_m are mapped to an answer bank $\{a_{m,1}, \dots, a_{m,n_m}\}$. The runner groups rows by `video_id`, parses timestamped subtitle cues, samples 16 chronological frames, and sends all movie questions in one request. The prompt requests concise English answers, cross-question consistency, and evidence support. Frames are JPEG encoded at a fixed maximum side and marked with timestamps. Long transcripts use a head-and-tail bound.

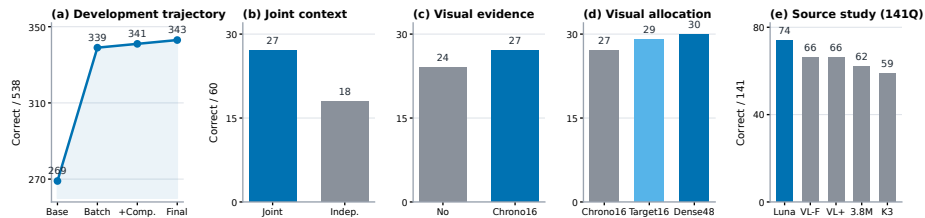


Figure 1. Evidence dashboard. (a) Development score trajectory. (b) Joint movie context versus independent question calls. (c–d) Visual configurations around the shared movie context. (e) Source comparison on a common 141-question movie-level overlap set. Each source receives the same evidence and question set: GPT-5.6-Luna, Qwen3-VL Flash, Qwen-VL Plus, Qwen3.8-Max, and Kimi-K3. Values are correct answers.

3 Inference Contract

The pipeline groups annotations, parses timestamped subtitles, samples the chronological frame pack, and sends one movie-level request. The answer bank uses the original question IDs and is converted directly to a two-column submission CSV.

4 Implementation

The runner accepts an OpenAI-compatible Responses endpoint. The conversion and audit utilities produce a complete `question_id,prediction` CSV with one answer for every requested ID.

Component	Configuration
Context	One movie, all questions jointly
Evidence	Timestamped transcript + 16 chronological JPEG frames
Frame encoding	Maximum side 512 px, JPEG quality 82
Generation	Temperature 0, high reasoning effort
Output	JSON bank, then <code>question_id,prediction</code> CSV

5 Results

The final SLoMO submission by *Math* ranked first with 66.36% accuracy and a 3.02 score. Figure 1 reports the mechanism evidence supporting the final system. Joint movie answering reaches 27/60, compared with 18/60 for independent calls, a gain of nine correct answers. The same visual budget with chronological frames reaches 27/60, while targeted and dense configurations reach 29/60 and 30/60. Panel (e) evaluates complete movie-level answer banks on a common 141-question overlap set. Every source receives the same evidence and questions. GPT-5.6-Luna reaches 74 correct answers, ahead of Qwen3-VL Flash, Qwen-VL Plus, Qwen3.8-Max, and Kimi-K3, and supplies the answer source used by the final system.